Lesson notes · for you

How things talk

IoT · Level 4 · Lesson 1

Materials
Paper and a pen for a table; Connected things at home or school: look at them, don't open them; A phone's settings, with permission

Learning goals

- Compare Bluetooth, Wi-Fi, mobile networks and LoRa by range, power and data.
- Follow a sensor reading from the device to an app, and say who handles it on the way.
- Choose a network for a device from what it needs.

Activity: Radio hunt

1. Walk round your home or school and list eight connected things: phones, a TV, the router, earbuds, a smartwatch, a bus card reader, a prepaid meter. Draw a table with five columns: device, your guess, what you found, what it sends, who might see it.
2. For each one, guess how it talks: Bluetooth, Wi-Fi, a mobile network, LoRa, or not connected at all. Think about how far it is from a router, and whether it has a SIM card.
3. Check your guesses: look at a phone's Bluetooth and Wi-Fi settings, read the label on the back (look for Wi-Fi, Bluetooth or a SIM slot), or find the manual online.
4. For each device, write what it sends (steps, video, location, meter readings) and who handles it on the way: the router's owner, the mobile company, the maker's cloud.
5. Circle the device whose data you'd most like to protect, and write one step you could take, such as checking that its app uses https or switching off a feature.

Only look at devices: don't open them, unplug them or change their settings without permission. Leave passwords and addresses out of your table and your photos.

Discussion

- Which device surprised you, and why?
- Why would a farm choose LoRa rather than Wi-Fi?
- Who handles your data on the way, and which of them do you trust?

Check yourself

- Choose a network for a rain gauge 3 km away, and say why.

Try it this week
Radio hunt: list five connected things at home, guess how each one talks (Bluetooth, Wi-Fi or a mobile network), then check its settings or its label.

Worksheet

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Name _____ Class _____ Date _____

1 **Aisha's family camera sends video to an app with no encryption. Who could watch the video on its way?**

- ◯ Anyone who handles it on the way
- ◯ Only Aisha's family: it's their camera
- ◯ Only the company that makes the app

2 **Match each device to the network that suits it best.**

A fitness band sending steps to a phone	• • • •	LoRa
A home camera streaming video	• • • •	Mobile network
A bus sending its location from across the city	• • • •	Bluetooth
A battery water-tank sensor 3 km away	• • • •	A memory card, collected weekly
A rain logger that nobody needs to read until Friday	• • • •	Wi-Fi

3 **Put a reading's journey in order, from the sensor to your phone.**

- ☐ A cloud server stores it
- ☐ The gateway passes it to the internet
- ☐ The sensor measures 28 °C
- ☐ An app on your phone shows it
- ☐ Its radio sends the reading to a gateway

4 **On a café's Wi-Fi, even when a website shows the padlock (https), the hotspot owner can still see which website you are visiting.**

- ◯ True
- ◯ False

5 **Compare what one send costs the battery.**

One Wi-Fi send: 120 mA for 2 seconds = _____ milliamp-seconds. One LoRa send: 40 mA for 0.4 seconds = 16 milliamp-seconds. So one Wi-Fi send uses as much battery as about _____ LoRa sends.

Word bank: 60 · 3 · 240 · 15

✎ **Think of one connected device you or your family use. What does it send, who handles that data on the way, and are you happy for all of them to see it?**

Check your answers

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1 **Anyone who handles it on the way**

Owning the camera doesn't hide its video.

2 **A fitness band sending steps to a phone — Bluetooth**
A home camera streaming video — Wi-Fi
A bus sending its location from across the city — Mobile network
A battery water-tank sensor 3 km away — LoRa
A rain logger that nobody needs to read until Friday — A memory card, collected weekly

Match the network to the job: close to a phone with tiny data, Bluetooth; lots of data where there's a router, Wi-Fi; moving across a city, mobile; far away on batteries, LoRa.

3 **Top to bottom: 4, 3, 1, 5, 2**

Sensor, gateway, internet, cloud, app. Each step is run by someone different, which is why it matters who can see the data on the way.

4 **True**

https scrambles the pages you read and what you type, but not the site's name: the hotspot owner still sees it, and how much data you use, as the lab shows.

5 **One Wi-Fi send: 120 mA for 2 seconds = 240 milliamp-seconds. One LoRa send: 40 mA for 0.4 seconds = 16 milliamp-seconds. So one Wi-Fi send uses as much battery as about 15 LoRa sends.**

120 × 2 = 240, and 240 ÷ 16 = 15. Sending by Wi-Fi costs about fifteen times as much charge, which is why battery sensors far from power use LoRa, or send less often.